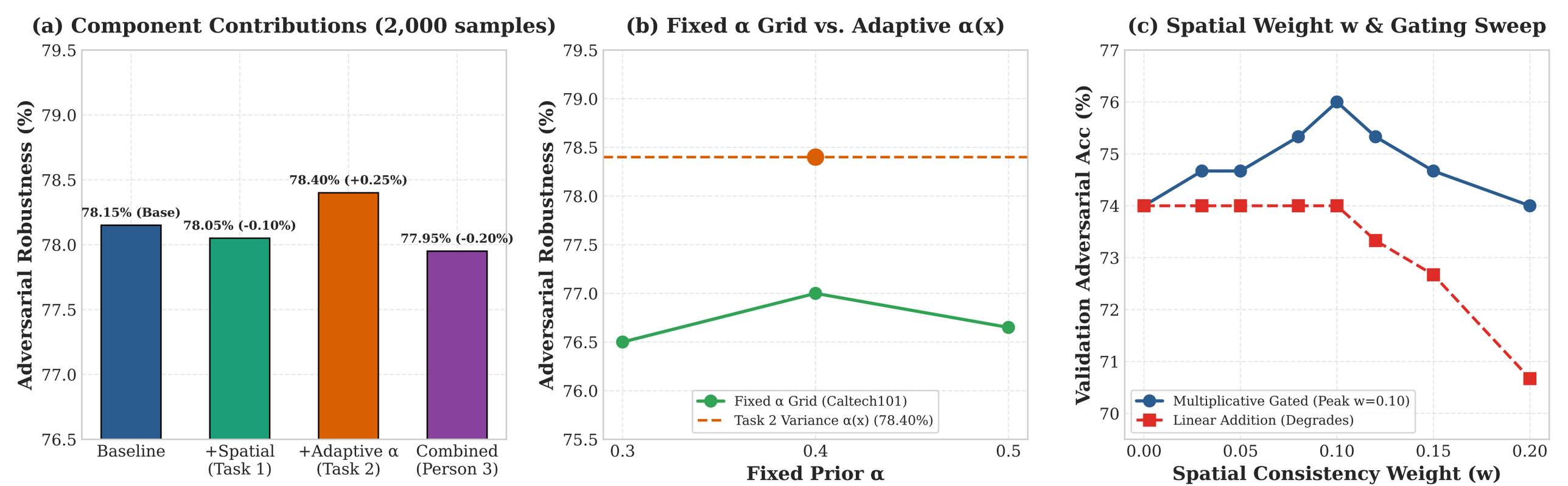
(a) Component Contributions (2,000 samples)
Adversarial Robustness (%)
79.5
79.0
78.5
78.0
77.5
77.0
76.5
78.15% (Base)
78.05% (-0.10%)
78.40% (+0.25%)
77.95% (-0.20%)
Baseline
+Spatial (Task 1)
+Adaptive α (Task 2)
Combined (Person 3)
(b) Fixed α Grid vs. Adaptive α(x)
Adversarial Robustness (%)
79.5
79.0
78.5
78.0
77.5
77.0
76.5
76.0
75.5
Fixed α Grid (Caltech101)
Task 2 Variance α(x) (78.40%)
0.3
0.4
0.5
Fixed Prior α
(c) Spatial Weight w & Gating Sweep
Validation Adversarial Acc (%)
77
76
75
74
73
72
71
70
Multiplicative Gated (Peak w=0.10)
Linear Addition (Degrades)
0.00
0.05
0.10
0.15
0.20
Spatial Consistency Weight (w)